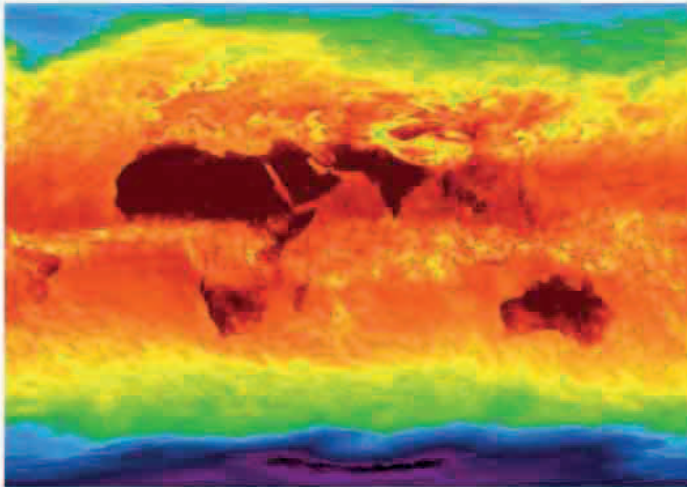


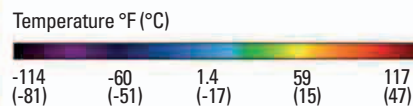
Oceans and continents

Continents warm up and cool down faster than oceans. This is because land has a lower heat capacity (the amount of energy it takes to change something's temperature) than water. Continental climates are therefore far more extreme than oceanic ones. These temperature differences also create local air currents, causing dramatic seasonal changes known as monsoons.



SLOW REACTION

Continents can get much hotter than oceans, as demonstrated in North Africa, Arabia, India, and Australia (all shown in dark red) on this satellite image, taken in April. The oceans never get quite as hot as this, nor as cold as the northern continents in winter. The oceans also stay warm for longer than the continents after the summer has ended, and stay cool for longer in spring.



COOL OR WARM WATERS

The coldest ocean water has a temperature of just below 32°F (0°C), and the sea ice at the North Pole is not much colder. By contrast, the continental ice of Antarctica can be colder than -58°F (-50°C). At the other end of the scale, tropical seas rarely get warmer than 86°F (30°C), but desert temperatures can soar above 140°F (60°C).

▼ WINTER SWIM

The water is much warmer than the air in the cold Russian winter. However, the water is still a bracing 32°F (0°C).

